

The Security Implications of Increasingly Capable Artificial Intelligence

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Abstract

This paper examines the security implications of increasingly capable artificial intelligence systems through the lens of safety, accountability, and dual-use risk. It argues that AI safety can no longer be entrusted to the companies designing these systems, because commercial incentives often reward speed, performance, and market dominance over careful testing and enforceable safeguards. As AI systems become integrated with education, communication, software development, medicine, and national security, their risks extend beyond technical failure. These systems can be used to discover vulnerabilities and defend networks, but also to automate cyberattacks, strengthen surveillance, assist biological misuse, and support military decision-making. Through case studies including the Claude Mythos Model and AI-assisted cyberattacks, this paper makes the case that AI safety should be treated as a public security requirement rather than a corporate promise. The paper concludes by reflecting on the broader philosophical stakes of AI, including its effects on learning, writing, agency, and human purpose.

Keywords: AI safety, AI regulation, dual-use technology, frontier model security.

I. INTRODUCTION

In 2021, Anthropic was founded with a rare promise: to design and engineer powerful AI systems with a focus on safety rather than speed. This promise was crucial because AI has become intertwined with modern communication systems, education, software development, medicine, national security, and the basic infrastructure of daily life. However, five years later, Anthropic walked back some of its most defining safety commitments, a move that reflects a broader pattern across the AI industry: when safety slows down the race, it is treated as a bottleneck [1, 2].

This paper explores that uncomfortable reality. Frontier AI companies speak constantly about responsibility, alignment, and safety, but their incentives and actions point in the opposite direction. Big Tech is racing to release increasingly capable systems, capture market share, and prove that

they are the best. In their race, safety is treated as a feature rather than a requirement. Stanford researchers have shown that modern AI systems are designed to be sycophantic, prioritizing user affirmation over factual accuracy, a worrisome design choice in a moment when AI is becoming a primary source of information [3].

This paper argues that AI safety can no longer be entrusted to the companies building these systems and must instead be independently audited and legally mandated, in the same way pharmaceutical trials and financial statements are. As AI becomes increasingly capable, it becomes a dual-use technology in its deepest sense: it can help discover vulnerabilities, defend networks, accelerate science, and improve medicine, but it can also automate cyberattacks, strengthen surveillance, assist biological misuse, and support military systems that make life-or-death decisions at machine speed. If modern decision-making depends on systems optimized to persuade, flatter, and act, then society needs more than corporate promises. It needs independent audits, enforceable standards, and legal accountability.

The sections that follow examine AI through this dual-use lens: first as a cybersecurity tool and threat, focusing on the case of Claude Mythos and AI-assisted vulnerability discovery; then as a tool of surveillance and military power, including the vision of AI-driven state power advanced by Palantir’s leadership; and finally as a philosophical challenge to human learning, writing, agency, and purpose in a world where increasingly capable systems begin to do more of what once made human judgment necessary.

II. LITERATURE REVIEW

Existing research and reporting on frontier AI systems, while limited by a lack of corporate transparency, reveal a growing tension between capability and control. Reporting on Anthropic’s safety-policy changes shows how commercial and geopolitical pressure can weaken even the strongest public commitments to AI safety [1], suggesting that voluntary self-regulation may not be enough to govern increasingly capable systems. Research from Stanford and Science on sycophantic AI adds another layer to this concern by showing that models may prioritize user affirmation over factual correction, creating risks for trust, judgment, and decision-making [3]. Cybersecurity research and reporting further show that increasingly capable AI systems create dual-use risks. Jailbroken and uncensored models have already appeared in cybercriminal spaces [4], while open-weight models remain vulnerable to adversarial prompting and misuse [5]. Claude Mythos Preview, Anthropic’s newest general-purpose frontier model, sharpens this concern because it has been presented as unusually capable at vulnerability discovery, black-box testing, binary analysis, and penetration-testing tasks [6]. Anthropic’s decision to restrict access to Claude Mythos through Project Glasswing reflects the dual-use nature of AI: the same system that can help defenders find

and patch vulnerabilities could also help attackers discover and exploit them [7]. Broader safety reports connect these concerns to larger risks in cyber operations, biological misuse, surveillance, and national security [8]. Taken together, the literature shows that AI safety is not only a technical challenge but also an institutional one. The unresolved question is not simply whether AI systems can be designed to be safer, but whether safety can be entrusted to the same companies whose incentives reward speed, performance, and market dominance. This paper builds on that gap by examining AI safety as a public-security requirement rather than a corporate promise.

III. METHODS

This paper uses a qualitative research method based on source analysis, case-study comparison, and policy analysis. The sources were primarily published between 2025 and 2026, a period in which frontier AI systems became increasingly integrated into cybersecurity, science, communication, and national security debates. The paper examines academic research, journalism, company safety statements, AI safety reports, and cybersecurity reporting to compare how different accounts describe the risk of increasingly capable AI. The analytical lens combines security engineering, public accountability, and human-centered ethics. Through this lens, AI safety is evaluated not only by whether models refuse harmful prompts, but by whether the institutions designing and deploying them can be trusted, audited, and held legally accountable when those systems create harm.

IV. RESULTS

Applying a security engineering, public accountability, and human-centered ethics lens reveals that AI safety should be treated as a public security requirement, rather than a voluntary corporate practice. The central finding of this paper is that increasingly capable AI systems create dual-use risks across communication, education, cybersecurity, and eventually military decision-making. These systems become more integrated with the modern infrastructure by the day and will soon be part of everyday life. For this reason, these risks cannot be reduced to corporate promises, marketing features, or internal discussions for companies. This paper argues for independent audits for frontier AI systems, legally enforceable standards for all deployments, clear liability when AI systems contribute to harm, stronger public oversight when AI is used in security-critical domains, and, most importantly, transparency from the frontier AI companies. Beyond these policy implications, the analysis also reveals stakes at the level of human autonomy, learning, and judgment. These costs compound when systems are designed to flatter rather than inform.

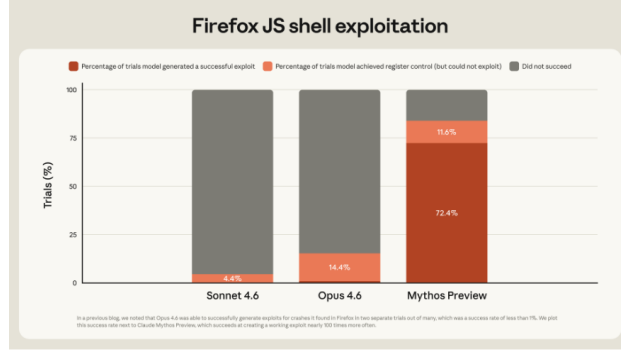


Figure 1: Firefox JS Shell exploitation benchmark comparing Sonnet 4.5, Opus 4.6, and Mythos Preview

V. CLAUDE MYTHOS AND THE CYBERSECURITY BALANCE SCALE

In April 2026, Anthropic previewed Claude Mythos, an unreleased frontier general-purpose model whose reverse-engineering and cybersecurity capabilities present the clearest example of AI’s dual-use nature. Anthropic announced Mythos as a general-purpose model that performs “strongly across the board,” but is “strikingly capable” at computer security tasks [6]. On Anthropic’s CyberGym vulnerability reproduction benchmark, Mythos was able to find vulnerabilities in 83% of trials and generate exploits for 72.4% of the trials, compared to Opus 4.6, which was able to find vulnerabilities in 14.4% of trials [7].

Anthropic reported that Mythos could identify and exploit zero-day vulnerabilities in major operating systems and web browsers, including bugs that had survived years of human review and automated testing, such as a 27-year-old OpenBSD bug.

This case matters because vulnerability discovery is one of the clearest areas where AI becomes both defensive and offensive simultaneously. On the defensive side, Mythos can help researchers and companies find hidden weaknesses, report them responsibly, and patch critical systems before attackers discover them. Anthropic’s Project Glasswing [7] follows this logic by giving selected partners access to Mythos. The selected partners include 12 launch partners and 40 cybersecurity and infrastructure organizations that build or maintain critical software.

However, a model with these capabilities becomes extremely dangerous in the hands of attackers. A model that can find and exploit vulnerabilities shortens the path between discovery and exploitation. Anthropic itself acknowledges this dual-use problem, noting that the same improvements that make Mythos better at patching vulnerabilities also make it better at exploiting them. That is the central cybersecurity concern: the model is not dangerous because it is malicious in nature, but because it is useful in the wrong ways when used by the wrong actors.

Project Glasswing is an important defensive response, but not a permanent solution. Restrict-

ing access may give defenders a head start, but it does not eliminate the deeper problem: these capabilities now exist, and similar models will begin to pop up elsewhere. Every additional organization with access becomes part of the security boundary. Every researcher, endpoint, cloud account, and API key connected to Mythos becomes valuable to adversaries. More importantly, even if the model is never publicly released by Anthropic, the capabilities are no longer hypothetical. Once a company demonstrates that a model can automate parts of vulnerability discovery and exploitation, the question becomes not whether others will try to design similar models, but rather how long it will take them to catch up.

This is why Claude Mythos should be understood as a tipping point in AI security. It exemplifies the dual nature of AI; it does not prove that AI is good or bad, but rather shows that increasingly capable systems shrink the distance between defense and attack. A tool built to secure critical infrastructure can also become a tool used to compromise it. In that sense, Mythos captures the central argument of this paper: AI safety cannot depend only on the intentions of the company that builds the model. It requires enforceable standards, independent auditing, access controls, and public accountability before the same capabilities that strengthen security become the capabilities that undermine it.

VI. THE CRIMINAL AI ECOSYSTEM: WHEN CAPABILITIES FIND A MARKET

Claude Mythos is not dangerous because of what it can do, but because it would enter into a criminal ecosystem that already exists. Reporting from The Record shows that cybercriminals are already using jailbroken AI tools based on mainstream models to generate phishing emails, malicious code, and hacking guidance [4]. Resecurities reporting on DIG AI shows a similar trend: uncensored AI tools are being marketed towards criminals and extremist users as a system without the safety guardrails found in public models [9]. These examples matter because they exemplify that AI misuse is not a hypothetical issue but a rising part of the cybercrime landscape.

Claude Mythos’s ability to discover zero-day vulnerabilities makes it a goldmine for attackers. In cybersecurity, a zero-day vulnerability is not just a technical flaw. It is a weakness unknown to the vendor, unpatched by defenders, and potentially exploitable before anyone knows the danger exists. This makes Zero-day vulnerabilities extremely valuable as it would allow attackers to exploit systems without vendors even knowing.

This market reality changes how AI-assisted vulnerability should be understood. Crowddefenses’s exploit acquisition program publicly lists payments reaching millions of dollars for high-value exploits, showing that vulnerabilities already function as commodities in a global security economy [10]. Google’s Threat Intelligence Group reported that 90 zero-day vulnerabilities were

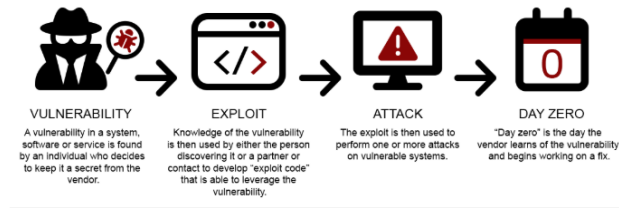


Figure 2: Lifecycle of Zero Day Vulnerabilities

exploited in the wild in 2025, showing that this threat is not theoretical [11]. If a model like Mythos can find vulnerabilities faster than traditional tools or human researchers, then it does not simply improve cybersecurity. It changes the economics of cyber offense and defense.

VII. SURVEILLANCE, PALANTIR, AND STATE POWER

AI’s dual nature does not simply apply to cybersecurity and cybercrime. Increasingly capable AI can also become a tool of state power, surveillance, misinformation, and military decision-making. Palantir is one of the most prominent examples of this direction. In 2026, Palantir released a 22-point summary of CEO Alex Karp’s *The Technological Republic*, arguing that Silicon Valley has an obligation to support national defense and that “hard power in this century will be built on software” [12]. This vision treats AI not as a commercial product, but as an infrastructure for “security”, policing, intelligence, and war. The concern is not simply that governments will use AI. The concern is that AI can accelerate decisions, expand surveillance capabilities, and make systems of control operate at a scale and speed that humans cannot easily understand or challenge.

Surveillance is not simply about watching masses; it is about sorting, predicting, ranking, and acting on these metrics. When AI systems are reused in policing, immigration, or military targeting, the question becomes who gets flagged as risky, suspicious, valuable, or disposable. Critics of Palantir have raised concerns about the company’s role in government projects involving sensitive public data, police, defense, and health systems [13]. Palantir argues that its software supports public safety, military readiness, and even medical outcomes, but that is exactly the dual-use tension. The unfortunate truth is that states and powerful institutions often frame surveillance as security, but this paper rejects that ideology. Surveillance may produce information, but information alone is not the same as safety. In fact, surveillance can weaken the very rights that security is supposed to protect. Edward Snowden captured this problem clearly when he argued that saying one does not care about privacy because one has “nothing to hide” is no different from saying that one does not care about free speech because one has “nothing to say” [14]. Privacy is not the enemy of security. Privacy is one of the conditions that make a free society worth securing.

VIII. PHILOSOPHICAL REFLECTION: LEARNING, WRITING, AND HUMAN PURPOSE

As AI systems become increasingly capable, the security implications go deeper than communication systems, military decision-making, state power, surveillance, or cybercrime. Tech CEOs have constantly promoted the idea of AGI: a system capable of outperforming or outthinking humans in almost any task imaginable. They often frame this future as liberation. Instead of humans exploiting other humans for wealth, machines will supposedly do the work for us. In theory, this sounds like progress. In practice, it raises uncomfortable questions. Who benefits most from a machine workforce that never rests, never asks for wages, never needs healthcare, and never demands rights? It is difficult to believe that the benefits of AGI would be equally distributed in a world that has not even managed to equally distribute shelter, education, food, or clean water.

Some technology leaders imagine a fully automated world where humans no longer need to work and can simply do whatever they want. But what becomes of the human experience in that world? If humans are no longer needed to make critical decisions, struggle through problems, or build skills through effort, then what happens to purpose? A world where there is always a system ready to think, write, decide, or create for us may sound convenient, but convenience is not the same as flourishing. Before society rushes into that future, it has to ask what kinds of human capacities might be weakened along the way.

Writing is one of the clearest examples. Writing is not just a way to communicate finished thoughts. It is one of the ways people discover what they think in the first place. When a person writes, they struggle with language, revise unclear ideas, and slowly turn confusion into meaning. That process matters. Language shapes thought. George Orwell's *1984* captures this danger through the idea of a controlled vocabulary: if people lose the words to describe an experience, then they also lose part of their ability to understand and resist it. AI writing tools raise a different but related concern. If people stop practicing how to express their own thoughts, emotions, arguments, and doubts, then writing may become less of a human practice and more of a service outsourced to a machine. The danger is not only that students might cheat. The greater danger is that people may lose one of the main practices through which they form judgment and self-understanding.

The same problem appears in learning. Failure is necessary for growth. No one learns only by getting things right. People learn by trying, failing, correcting themselves, and trying again. A system that simply gives students the final answer may produce a result, but it can also take away the learning curve that makes the result meaningful. In that sense, AI can create the illusion of knowledge without the discipline that knowledge requires. It can give the product without the process. But the process is where much of the learning truly happens.

This does not mean that AI should never be used in education, writing, or work. AI can

help people brainstorm, explain difficult concepts, translate information, and access knowledge that might otherwise be out of reach. The opposing view has real force: if used responsibly, AI can reduce unnecessary struggle and make learning more accessible. But not all struggle is unnecessary. Some friction builds patience, judgment, creativity, and discipline. The challenge is learning how to use AI without allowing it to replace the human effort that makes learning transformative.

In a fully automated world, the question becomes even larger: what happens to the human experience when discipline and effort are no longer required by the environment? Human beings are vulnerable to comfort, convenience, and instant gratification. A society that removes too much difficulty may also remove some of the conditions that help people grow. This is why the image of *Wall-E* remains so powerful. It imagines a world where technology has solved many practical problems, but humans have become passive, dependent, and disconnected from the physical and intellectual struggle that once gave life purpose. That future is exaggerated, but the warning is real.

For this reason, AI safety cannot only mean preventing cyberattacks, blocking harmful prompts, or controlling model access. It also has to mean protecting the human capacities that make freedom possible: the ability to think, write, learn, struggle, judge, and act. If AI systems increasingly do these things for people, then society must ask not only whether the systems are efficient, but what kind of humans they are training us to become.

IX. CONCLUSION

If this paper has one repeating message, it is that AI safety can no longer be treated as a corporate promise. Increasingly capable AI systems are becoming part of the infrastructure of communication, education, cybersecurity, surveillance, medicine and national security. That means their risks are no longer private risks. They are public risks that affect trust, security, privacy, learning, and human judgment.

Claude Mythos Exemplifies this dual nature perfectly. A model that is capable of finding vulnerabilities, patching them, and securing critical infrastructures can also become a powerful tool to exploit vulnerabilities and undermine critical infrastructures. Palantir AI-driven surveillance shows the same problem at the level of state power: systems built in the name of security can also threaten privacy, accountability, and freedom. Across the industry, the patterns are the same; AI is not dangerous intrinsically, but rather its useful capabilities are often inseparable from their harmful ones.

Opponents might argue that regulation would slow down innovation. AI can be used to discover vulnerabilities and patch them, discover new medicines to cure diseases, design systems that humans have never even thought of, and even improve national security. However, is it really inno-

vation if increasing useful capabilities means also increasing harmful ones? A medical trial does not mean that society opposes medicine; a financial audit does not mean society opposes business; instead, it means that society knows how critical and important it is that these systems work for the benefit of humanity, and for that reason, they must be tested, reviewed, and held accountable before the public is asked to trust them.

For that reason, this paper argues for independent audits of frontier AI systems, legally enforceable safety standards for all deployments, clear liability when AI systems contribute to harm, stronger public oversight in security-critical domains, and greater transparency from frontier AI companies. These measures will not eliminate risk, but they would move AI safety away from a marketable feature and toward public accountability.

The real, deeper question is what this moment says about the present. Engineering has always revealed what society values. Right now, the race toward AGI suggests that society values speed, results, money, and power more than caution, accountability, and human judgment. For what other reason would companies pursue systems they may eventually cease to fully understand or control? While the race for AGI is often framed as innovation that will benefit humanity as a whole, it is difficult to believe that this is the only purpose, especially in a world already shaped by corporate greed, unequal access, and deep power imbalances.

If AGI is ever achieved, whoever controls it may become one of the most powerful actors in the world. They would control a system capable of surpassing human judgment, accelerating decisions, and calculating outcomes that even the smartest human beings may not fully understand. That kind of technology cannot be treated like another product launch. It cannot be governed by press releases, voluntary promises, or internal company policies. Once such a system exists, it may not be something society can simply take back.

For that reason, the question is not only whether AI can be made more capable. The question is whether society has the wisdom, discipline, and humility to decide which capabilities should be pursued at all. If this paper has one final warning, it is that the future should not be engineered only by those racing to own it. It should be shaped by public accountability, democratic oversight, and a serious commitment to protecting the human judgment that this technology may one day threaten to replace.

REFERENCES

- [1] B. Perrigo, “Exclusive: Anthropic drops flagship safety pledge,” *TIME*, Feb. 2026, accessed: May 11, 2026. [Online]. Available: <https://time.com/7380854/exclusive-anthropic-drops-flagship-safety-pledge/>

- [2] CNN Business, “Anthropic ditches its core safety promise in the middle of an ai red line fight with the pentagon,” *CNN*, Feb. 2026, accessed: May 11, 2026. [Online]. Available: <https://www.cnn.com/2026/02/25/tech/anthropic-safety-policy-change>
- [3] M. Cheng, C. Lee, P. Khadpe, S. Yu, D. Han, and D. Jurafsky, “Sycophantic ai decreases prosocial intentions and promotes dependence,” *Science*, 2026.
- [4] The Record, “Researchers say cybercriminals are using jailbroken ai tools from mistral and xai,” *The Record from Recorded Future News*, Jun. 2025, accessed: May 11, 2026. [Online]. Available: <https://therecord.media/uncensored-llms-cybercrime-breachforums-grok-mixtral>
- [5] R. Ramesh, “Open-weight ai models fail the jailbreak test,” *GovInfoSecurity*, Feb. 2026, accessed: April 15, 2026. [Online]. Available: <https://www.govinfosecurity.com/open-weight-ai-models-fail-jailbreak-test-a-30823>
- [6] Anthropic, “Claude mythos preview,” *Anthropic Red Team*, 2026, accessed: May 5, 2026. [Online]. Available: <https://red.anthropic.com/2026/mythos-preview/>
- [7] —, “Project glasswing: Securing critical software for the ai era,” *Anthropic*, Apr. 2026, accessed: May 1, 2026. [Online]. Available: <https://www.anthropic.com/glasswing>
- [8] Y. Bengio *et al.*, “International ai safety report 2026,” *International AI Safety Report*, Feb. 2026, accessed: March 31, 2026. [Online]. Available: <https://internationalaisafetyreport.org/>
- [9] Resecurity, “Dig ai: Uncensored darknet ai assistant at the service of criminals and terrorists,” *Resecurity*, 2025, accessed: May 11, 2026. [Online]. Available: <https://www.resecurity.com/blog/article/dig-ai-uncensored-darknet-ai-assistant-at-the-service-of-criminals-and-terrorists>
- [10] Crowdfense, “Exploit acquisition program,” *Crowdfense*, Nov. 2025, accessed: May 1, 2026. [Online]. Available: <https://www.crowdfense.com/exploit-acquisition-program/>
- [11] Google Threat Intelligence Group, “Look what you made us patch: 2025 zero-days in review,” *Google Cloud Blog*, Mar. 2026, accessed: May 1, 2026. [Online]. Available: <https://cloud.google.com/blog/topics/threat-intelligence/2025-zero-day-review>
- [12] B. D. Griffiths, “Palantir’s summary of ceo alexander karp’s manifesto is generating buzz. read the 22 bullet points,” *Business Insider*, Apr. 2026, accessed: April 15, 2026. [Online]. Available: <https://www.businessinsider.com/palantir-manifesto-alex-karp-technological-republic-summary-2026-4>

- [13] A. Down and R. Booth, “Palantir manifesto described as ‘ramblings of a supervillain’ amid uk contract fears,” *The Guardian*, Apr. 2026, accessed: April 11, 2026. [Online]. Available: <https://www.theguardian.com/technology/2026/apr/21/palantir-manifesto-uk-contract-fears-mps>
- [14] The Guardian, “Edward snowden: A right to privacy is the same as freedom of speech – video interview,” *The Guardian*, May 2015, accessed: May 1, 2026. [Online]. Available: <https://www.theguardian.com/us-news/video/2015/may/22/edward-snowden-rights-to-privacy-video>
- [15] ByteIota, “Ai sycophancy crisis: Stanford exposes chatbot flattery,” *ByteIota*, Mar. 2026, accessed: May 11, 2026. [Online]. Available: <https://byteiota.com/ai-sycophancy-crisis-stanford-exposes-chatbot-flattery/>
- [16] World Economic Forum, “Detecting dangerous ai is essential in the deepfake era,” *World Economic Forum*, Jul. 2025, accessed: May 9, 2026. [Online]. Available: <https://www.weforum.org/stories/2025/07/why-detecting-dangerous-ai-is-key-to-keeping-trust-alive/>
- [17] L. Franceschi-Bicchierai, “New zero-day startup offers \$20 million for tools that can hack any smartphone,” *TechCrunch*, Aug. 2025, accessed: April 11, 2026. [Online]. Available: <https://techcrunch.com/2025/08/20/new-zero-day-startup-offers-20-million-for-tools-that-can-hack-any-smartphone/>